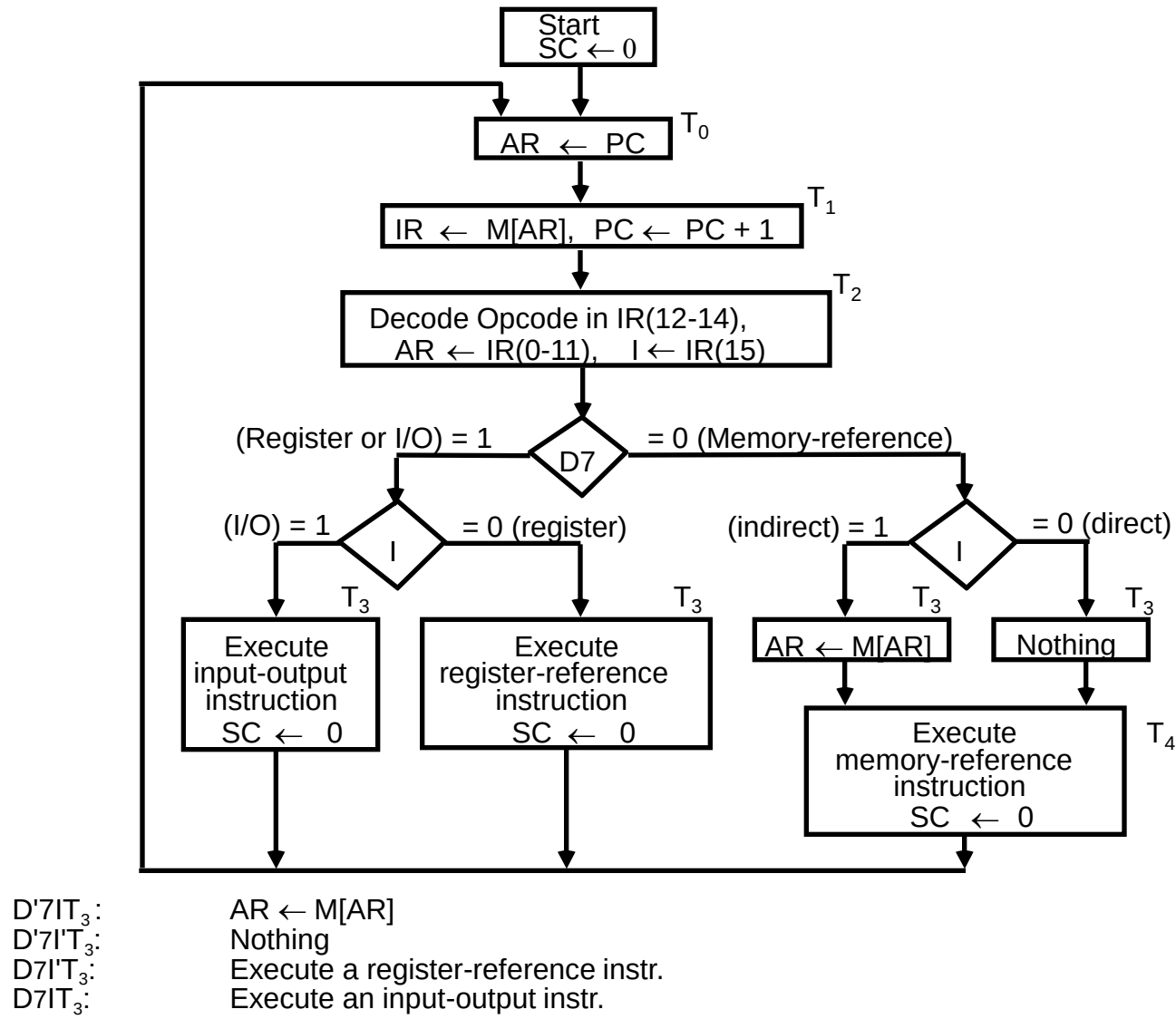


DETERMINE THE TYPE OF INSTRUCTION



REGISTER REFERENCE INSTRUCTIONS

Register Reference Instructions are identified when

- $D_7 = 1, I = 0$
- Register Ref. Instr. is specified in $b_0 \sim b_{11}$ of IR
- Execution starts with timing signal T_3

TABLE 5-3 Execution of Register-Reference Instructions

$D_7I'T_3 = r$ (common to all register-reference instructions)

$IR(i) = B_i$ [bit in $IR(0-11)$ that specifies the operation]

	r :	$SC \leftarrow 0$	Clear SC
CLA	rB_{11} :	$AC \leftarrow 0$	Clear AC
CLE	rB_{10} :	$E \leftarrow 0$	Clear E
CMA	rB_9 :	$AC \leftarrow \overline{AC}$	Complement AC
CME	rB_8 :	$E \leftarrow \overline{E}$	Complement E
CIR	rB_7 :	$AC \leftarrow \text{shr } AC, AC(15) \leftarrow E, E \leftarrow AC(0)$	Circulate right
CIL	rB_6 :	$AC \leftarrow \text{shl } AC, AC(0) \leftarrow E, E \leftarrow AC(15)$	Circulate left
INC	rB_5 :	$AC \leftarrow AC + 1$	Increment AC
SPA	rB_4 :	If $(AC(15) = 0)$ then $(PC \leftarrow PC + 1)$	Skip if positive
SNA	rB_3 :	If $(AC(15) = 1)$ then $(PC \leftarrow PC + 1)$	Skip if negative
SZA	rB_2 :	If $(AC = 0)$ then $PC \leftarrow PC + 1$	Skip if AC zero
SZE	rB_1 :	If $(E = 0)$ then $(PC \leftarrow PC + 1)$	Skip if E zero
HLT	rB_0 :	$S \leftarrow 0$ (S is a start-stop flip-flop)	Halt computer

